



Faults in parallel: meshing and cutting across the partition

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A fault in a geodynamic model is a thin object in a large mesh. It needs its own resolution — a band a few elements across, along a trace that wanders where the geology says it wanders — and that band has to be built into a mesh whose cells are elsewhere a hundred times larger. In serial this is mesh surgery: cut a cavity out of the background mesh, mesh the fault's own layer, sew the two together, and if the fault is to slip rather than merely deform, duplicate the nodes along its mid-surface so the two sides can move past one another.

In parallel none of that is local. The mesh is distributed before the fault is placed, and the partitioner divides it to balance the cell count, knowing nothing about where the fault will go. Worse, the partitioner is *attracted* to the fault: the region is about to be refined, so a balanced partition tends to put a boundary through exactly the part of the mesh the surgery needs to see whole.

This note is about doing the surgery anyway. It is the parallel half of a larger problem; the serial constructions are assumed and described only far enough to say what breaks.

1. WHAT HAS TO BE TRUE FOR THE SURGERY TO WORK

Two operations, two different requirements.

Placing the band. A cavity is cleared from the background mesh, the fault's own layer is meshed to its own resolution, and the space between the layer's outer skin and the cavity's ring is filled. The fill is a triangulation of a region bounded by two curves, and it has to be *one* region: the mesher is handed a closed outline and returns cells inside it. A cavity that straddles a partition boundary is two half-regions on two processors, neither of which is a closed outline.

Cutting the mid-surface. The nodes along the fault are duplicated and the cells on one side are rewired to the copies, so that the finite element spaces carry a jump. Deciding which side a cell is on means walking the fan of cells around each node of the fault, in order. A node on the partition boundary has part of its fan on one processor and part on another, and neither can complete the walk.

Both requirements have the same shape: a local construction that assumes it can see a whole neighbourhood, applied to a mesh that has been cut into pieces without regard for that neighbourhood.

2. THE OBVIOUS ANSWER, AND WHAT IT COSTS

The obvious answer is to move the fault to one processor. Mark the cells touching the fault, plus a layer, and reassign them all to the processor that already owns most of them. That processor then holds the whole neighbourhood, does the surgery, and every processor rebuilds its part of the result. The requirement is met by construction.

It works, and for a single fault in a modest mesh it is the right thing to do. The cost appears in two places. The moved region is thin, so the imbalance is bounded by the fault's own size rather than by the refined band — but for a network of faults spread across the domain, moving them all to one processor moves a great deal, and the processor that receives them is left with a mesh that is not balanced against anybody else's. On a test with a fault network in a unit box, the gather moved 91% of the fixture.

The second cost is subtler and matters more. Because everything is gathered before the surgery, the *answer* is a serial answer computed on one processor and then redistributed. Nothing is learned about whether the construction is partition-independent, because it never runs on more than one partition.

3. LEAVING THE SEAM ALONE

A cheaper answer is to stop short. Each processor carves its own part of the cavity but stops one cell before the partition boundary, and the band's cells there are left as background material — a **ligament** — which the fault's rheology is painted onto but which the cut does not pass through. Nothing moves. A fault that crosses a processor twice is cut as two pieces, each with its own ends.

This is a genuine option for a fault represented as a weak zone, where the ligament simply carries a slightly coarser weak band across the boundary. On a long fault crossing one seam it costs a few tenths of a percent in the peak slip rate, and the cells stay balanced.

It is not an option for a fault represented as a slipping cut. A cut that stops short leaves a **pinned tip** at every crossing, and a tip is where slip goes to zero. The fault is welded to itself at the partition boundary. Measured on the same long fault, the cut lost 13% of its peak slip per crossing, and — the diagnostic that settles it — the loss did not respond to the weak-plane viscosity painted on the ligament. No rheology can free a pinned tip.

4. MESHING THE BAND THROUGH THE SEAM

The alternative is to admit that the cavity straddles the boundary and to mesh the band on both sides, so that the two processors' meshes agree along a curve that lies *inside* the band.

The construction that works turned out to be simpler than the one first sketched. There is no negotiated interface surface and no realignment of the partition. Instead:

- **Ownership is decided by the cavity.** Every cell of the fault's own layer belongs to the processor whose cleared cells contain its centroid; every vertex of the layer, and every edge of its outer skin, likewise. Three global arrays, one reduction each. The boundary between the two processors' layer cells is then a chain of the layer's own edges — the seam *inside* the band — and needs no construction at all.
- **Deletion at the seam is one decision.** A vertex of the background mesh that lies on the partition boundary is removed only if the band actually reaches it, never merely because it falls inside the carve's clearance, and the decision is reconciled between the processors that hold it, so both delete it or neither

does. A boundary that runs *beside* the band keeps its edges, and the fills on either side keep their common boundary there.

- **The fill's outline is a graph.** Each processor's fill is bounded by its own ring edges that were not cut, its own share of the skin, and one connector from each surviving end of a cut span to the nearest free end of its skin runs. Every vertex then has exactly two edges, and walking the graph gives the outlines, with enclosed loops filled as holes. A junction the boundary crosses twice, a band the boundary skirts, and a fault ending near the boundary all come out of the same walk.
- **The new shared points enter the mesh's own bookkeeping.** Vertices of the layer that more than one processor creates are the same point, owned by the lowest-numbered of them, and they are registered as such before the mesh's edges and faces are built — an ordering that matters, because the alternative leaves each processor's faces oriented however its own local construction chose them.

In serial the whole path reduces to the previous one exactly.

5. CUTTING THROUGH THE SEAM

With the band meshed through the boundary, the cut can follow. The obstruction is the fan walk, and the answer is to give every processor the same fault.

The fault's facets are exchanged as pairs of *global* vertex identities — the owning processor and its local index, which the mesh's own sharing structure already provides — together with their positions. Every processor assembles the same paths from the union, so a vertex on the partition boundary has the same two neighbours everywhere, and the walk order, which decides which side is which, is one decision taken identically rather than several taken locally.

At a vertex not on the boundary the fan is walked as before. At a vertex on it, the fan cannot be walked, but it does not need to be: the two fault facets there are edges of cells, so every local cell lies wholly within one of the two angular sectors those facets bound, and the sector containing its centroid is its side. Either sector may be empty, which is what happens when the partition boundary touches the fault without crossing it.

The duplicate node is then created on *every* processor that holds the original, owned where the original is owned. That last clause is not bookkeeping fussiness: the two nodes of a coincident pair carry the constraint that couples them, and a pair split across processors would put that constraint in nobody's diagonal block.

One quantity has to be repaired afterwards. The interface condition needs a normal at each pair of nodes, and it is accumulated from the fault facets around the node, weighted by facet length. At a node on the partition boundary a processor holds one of the two facets. Rather than exchange normals at solve time — a collective call inside a routine that can fail locally, which is how deadlocks are made — the split records the whole sum, which it already has from the global fault, and the accumulator uses it in place of its own half.

6. WHAT IT MEASURES

The test is not that the parallel run completes. It is that the parallel run gives the serial answer while the cells stay balanced, because the whole point of not gathering is that nothing was quietly serialised.

A single fault of 35 elements across a unit box, crossed once by the partition boundary at two processors and twice at three, solved as a frictionless slipping cut:

processors	cells per processor	coincident pairs	peak slip rate
1	2264	69	0.5094
2	1212 / 1042	69	0.5094
3	785 / 890 / 571	69	0.5094

The pair count is the serial count, including the pair at the crossing itself; the slip rate is the serial one to four figures; and no processor holds more than 60% of the cells. Where the ligament construction lost 8 to 15% per crossing, this loses nothing that can be measured. A three-strand network crossed mid-fault gives the same result on every strand.

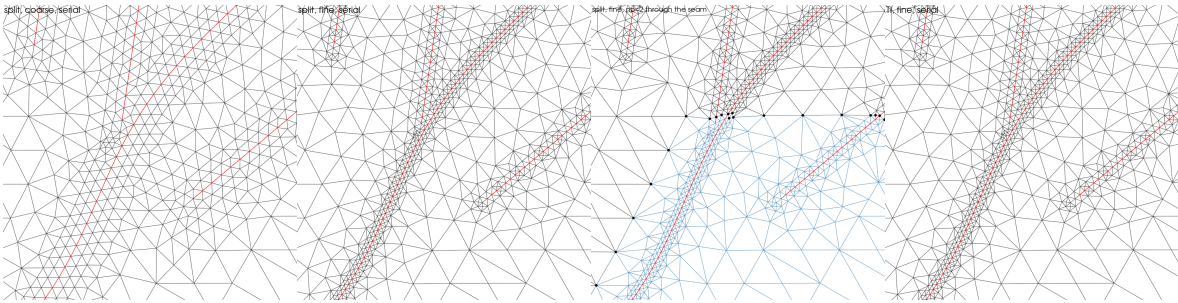


Figure 1: The mesh where the partition boundary crosses the fault. The band keeps its own resolution through the crossing; in the parallel case (third panel) the boundary between the two processors' cells runs along the band's own edges, and the vertices both processors hold are marked.

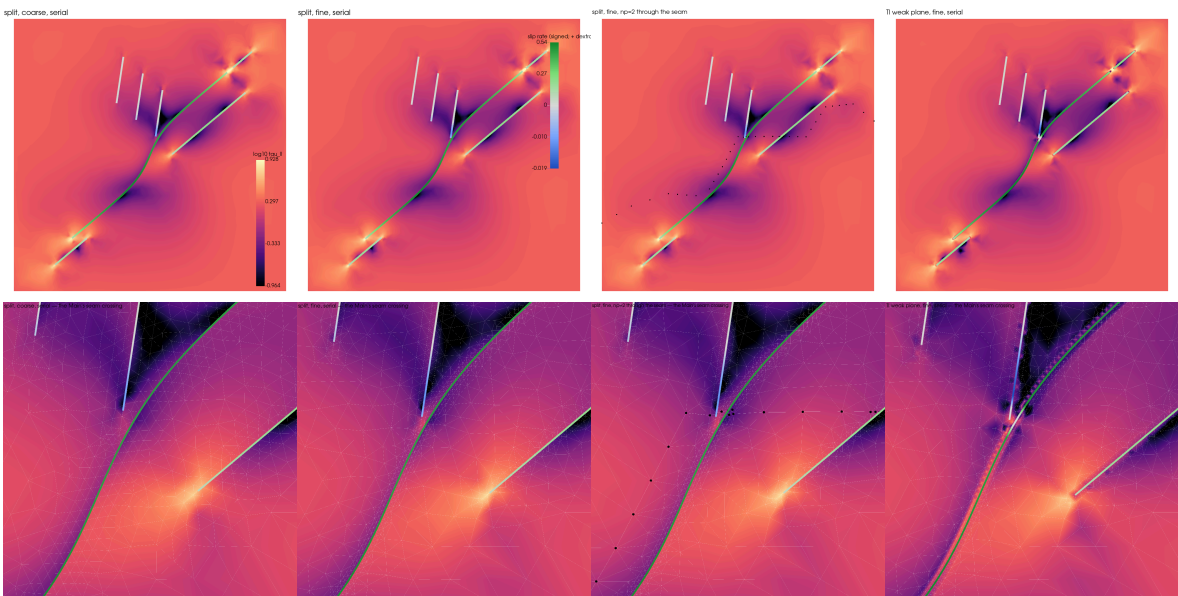


Figure 2: Stress and slip for the same network: the cut in serial at two resolutions, the cut through the partition boundary at two processors, and the weak-band representation for comparison. The traces carry the signed slip rate, positive for right-lateral.

7. WHAT IS REFUSED

Three configurations are declined rather than approximated, and saying so plainly is more useful than a construction that silently produces something else.

A partition boundary that runs along the fault. If the boundary follows the fault's own facets rather than crossing them, the two cells of a fault facet belong to different processors, and the two copies of that facet after the cut would too. There is nothing to build the constraint in. The placement now also keeps both cells of a fault facet with a single processor, which is what closed a case where a three-processor partition had put one such facet on the boundary.

A fault tip exactly on the partition boundary, beside a junction. The fill's outline walk doubles back on itself there and the resulting outline crosses itself.

A junction within a band width or two of the boundary. The connector that closes the fill's outline runs from a ring end to the nearest free end of a skin run, in a straight line, with no check that the line is clear — and next to a junction it can cross the bump that another strand's tip margin makes on the skin. The fix is a retarget that both processors agree on, which needs an exchange they do not currently have.

The last two are refusals rather than failures because a self-crossing outline is detected before the mesher sees it. That check was worth adding on its own account: handed a self-crossing outline, the mesher does not refuse, it spins, and a run that has stopped producing output is considerably harder to diagnose than one that has stopped with a message.

8. WHERE THIS LEAVES THINGS

For a fault represented as a weak band, or as a slipping cut, in two dimensions: the band is meshed through the partition boundary, the cut passes through it, nothing is gathered, and the answer is the serial answer. For three dimensions the constructions carry over one dimension up — the fault is a surface rather than a curve, its cut needs a global patch rather than a global chain, and the side rule at a shared rim vertex comes from the patch's own face normals — but that is not built, and the three-dimensional thin volume still gathers.

The refusals above are the honest boundary of the two-dimensional case, and two of the three are about a fault junction sitting close to a partition boundary, which is a coincidence of the decomposition rather than anything a modeller chooses. A partition-steering answer — deciding where the boundary may fall before the mesh is divided, rather than working around where it fell — would retire all three, and is the obvious next thing to try.